ICTcotas User Manual

TradingView indicator (Pine Script v6, overlay) by franjballest. This manual describes the script as it is written in `ICTcotas.txt`.

1. Overview

ICTcotas draws higher timeframe PD arrays (price delivery arrays) on your chart so you can trade lower timeframes against daily, weekly and monthly reference levels. It draws:

Element	Source candles	Drawn as
Fair Value Gap (FVG) and Inversion FVG (IFVG)	Daily, weekly, monthly	High line, low line, CE line, optional shaded rectangle, labels
Volume Imbalance (VI)	Daily, weekly, monthly	High line, low line, CE line, optional shaded rectangle, labels
Wick CE (upper and lower)	Daily, weekly, monthly; optionally 1H and 4H	One CE line with a label
FVG by timeframe (and its IFVG)	15m, 1H, 4H, D, W, M (your choice)	Outline box, CE line, formation marker, one label

Every level extends from the candle where it formed to a few bars to the right of the current bar. When a level is mitigated (according to the rule you choose) it is removed from the chart. FVGs that are closed through can instead turn into IFVGs and keep being drawn with the opposite direction.

The indicator computes no entry signals and has no alerts. It is a map of levels, not a strategy.

The two layers

- **Timeframe levels** (sections General through Wick CE in the settings): FVGs, VIs and wick CEs of the daily (D), weekly (W) and monthly (M) candles, drawn as horizontal lines. Weekly lines are thicker than daily ones and monthly lines thicker still.
- **FVGs by timeframe** (the last settings section): FVGs of 15m, 1H, 4H, D, W and M candles drawn as boxes, each with its own colors. This layer is on by default with 1H and 4H selected. It can also add wick CE lines for 1H and 4H.

2. Quick start

1. Add ICTcotas to the chart (Indicators, then search for ICTcotas in your scripts).
2. Use an intraday execution chart for NQ or DAX, for example 1m, 3m, 5m or 15m. ICTcotas only shows levels of timeframes higher than the chart, never the chart’s own: daily levels need a chart below 1D, weekly below 1W and monthly below 1M.
3. Leave the defaults to start: D, W and M levels on, FVG / VI / wick CE on, 8 of each kept per timeframe, levels removed when fully filled by a wick, FVGs turning into IFVGs on a close through them, and 1H and 4H FVG boxes.

4. If the chart looks crowded, untick the kinds you do not use (for example Draw VIs), or reduce **keep**.
5. Scroll the chart to the left once or twice if weekly or monthly levels are missing: the script only finds levels as higher timeframe candles complete inside the loaded chart history (section 9).

Recommended setups

Use	Chart	Suggested settings
Intraday NQ / DAX execution	1m to 5m	Defaults. Optionally tick 15m in the by-timeframe section, or set **plus the next higher** to 2
Intraday bias and planning	15m	Defaults; 1H and 4H boxes give the intraday PD arrays, lines give D/W/M
Swing context	1H or 4H	Defaults; tick D, W or M in the by-timeframe section only if you want box versions (they duplicate the D/W/M lines)

Platform time zone

The time zone of the chart does not change any level. Daily, weekly and monthly candles are the exchange's own candles for the symbol (for CME NQ the trading day starts at 18:00 New York time; for Eurex DAX it follows the Eurex session). Setting the chart to New York time (UTC-5 / UTC-4) is useful only for reading the chart in ICT terms (midnight open, kill zones); the indicator does not depend on it.

3. Reading the chart

Colors and direction

Each timeframe has a bull and a bear color. Bullish levels (support, expected to hold price from below) use the bull color; bearish levels (resistance) use the bear color.

Timeframe	Bull (default)	Bear (default)
Daily lines	green	red
Weekly lines	teal	orange
Monthly lines	blue	purple
15m boxes	lime	maroon
1H boxes	green	red
4H boxes	teal	orange
D boxes	blue	fuchsia
W boxes	navy	purple

Lower wicks count as bullish and upper wicks as bearish. An IFVG takes the color of its new (flipped) direction: a bullish daily FVG that is inverted turns red.

Line thickness

For timeframe levels the width is the base width of the kind (default 1) plus an extra amount per timeframe step: daily +0, weekly +1, monthly +2 with the default **extra line width per timeframe step** of 1. Thicker line means higher timeframe.

Labels

Timeframe-level labels sit just to the right of each line end and read:

`<timeframe> <name><sign> <part>`

Example	Meaning
`D FVG hi`	Daily FVG, upper edge
`W FVG CE`	Weekly FVG, midpoint
`M IFVG lo`	Monthly FVG that has been inverted, lower edge
`D VI hi`	Daily volume imbalance, upper edge
`W UW CE`	CE of the longest upper wick in a weekly wick group
`D LW CE`	CE of the longest lower wick in a daily wick group
`1H UW CE`	Upper wick CE from 1H candles (by-timeframe wick option)
`D FVG+ hi`	Same as `D FVG hi` with **direction (+/-)** on: + bullish, - bearish

When the hi, lo and CE labels of a level would sit too close to read (closer than **one label when hi / lo / CE closer than x daily ATR**, 0.05 of the daily ATR(14) by default), the level gets a single label without the part, `D VI`, at its CE instead of three labels on top of each other. The lines are still drawn.

The sign is added only to FVG, IFVG and VI labels, never to wick labels. All words (D, W, M, FVG, IFVG, VI, UW, LW, hi, lo, CE) can be changed in the settings. Box labels of the by-timeframe layer are described in section 7.

4. Fair Value Gaps and IFVGs (D / W / M)

What you see

A daily, weekly or monthly FVG is drawn as three horizontal lines, its high (solid), low (solid) and CE (dotted), over a lightly shaded rectangle (88 percent transparent) in the timeframe color. The lines start at the middle candle of the three-candle pattern and run to the right of the current bar.

What it is

A Fair Value Gap is a three-candle pattern where the wicks of the first and third candle do not overlap, leaving a price range that was delivered in one direction only.

- **Bullish FVG**: the low of the newest candle is above the high of the oldest candle. The gap runs from the oldest candle’s high up to the newest candle’s low.

- **Bearish FVG**: the high of the newest candle is below the low of the oldest candle. The gap runs from the newest candle's high up to the oldest candle's low.

The **CE** (consequent encroachment) is the midpoint of the gap.

With **extend FVG range to include VIs with candle 2** on (default), the FVG is widened when the first or third candle also has a body gap with the middle candle: a volume imbalance, or a gap when the wicks do not overlap either (at least the VI **min size**). The FVG then covers the gap plus that body gap, and the CE is recomputed. The VI itself is still drawn separately in the Volume Imbalances layer.

Inversion (IFVG)

With **FVG becomes IFVG on a close through it** on (default), a candle of the FVG's own timeframe that closes beyond its far edge inverts it:

- a bullish FVG inverts when a candle of its timeframe closes below its low and becomes a bearish IFVG (resistance);
- a bearish FVG inverts when a candle of its timeframe closes above its high and becomes a bullish IFVG (support).

The label changes from FVG to IFVG and the color switches to the opposite direction. While this option is on, a normal (not inverted) FVG is never removed by the mitigation rule; it only stays or inverts. The IFVG, once created, is removed by the mitigation rule in General.

Inversion is tested on the closes of the FVG's own timeframe, never on the chart's closes. A daily FVG on a 5m chart inverts only when a daily candle closes through it; 5m candles closing through it change nothing. The flip shows on the first chart candle after that daily candle closes.

Overlapping FVGs

When several FVGs of the same timeframe and direction overlap, **overlapping FVGs** decides what stays (IFVGs are never touched by this rule):

Option	Effect
Most recent (default)	A new FVG removes every older one it overlaps
Enclosing	If an existing FVG already contains the new one, the new one is discarded. Otherwise the new one replaces all older ones it overlaps
Keep all	Every FVG is drawn

How to use it

- Bullish FVGs below price are areas where a retracement can be expected to find support; the CE is the classic ICT reaction point and the low is the line in the sand.
- Bearish FVGs above price are resistance; look for rejections at the low edge or the CE.
- An IFVG is an FVG that failed. ICT traders use it as support or resistance in the new direction, often as a retest entry after a displacement through it.
- Weekly and monthly gaps are frequently the draw on liquidity for the week or month; daily gaps are the usual intraday targets and reaction zones.

5. Volume Imbalances (D / W / M)

What you see

Like FVGs: high and low lines (solid), a dotted CE line and a shaded rectangle, labelled `VI`.

What it is

A volume imbalance is a gap between the bodies of two consecutive candles while their wicks still overlap (so it is not a full gap).

- **Bullish VI**: the bottom of the newest candle's body is above the top of the previous candle's body, and the newest candle's low is not above the previous candle's high. The range is between the two bodies.
- **Bearish VI**: the mirror image.

The gap must be at least **min size** price points.

How to use it

VIs act like small FVGs: price often returns to the body gap and reacts at its edges or CE. On the weekly and monthly charts a VI frequently forms at the open of a new candle (weekend or month-start gaps in the body), which makes them useful targets on Mondays and the first days of a month.

6. Wick CE (D / W / M, plus optional 1H / 4H)

What you see

A single dashed line at the midpoint of a candle wick, labelled `UW CE` (upper wick) or `LW CE` (lower wick). There are no high, low or fill for wick levels.

What it is

A wick counts when it is at least **min wick %** of the candle's full range (default 30 percent). Consecutive candles with a qualifying wick on the same side form a group, and only the CE of the longest wick of the group is drawn. A single qualifying wick is its own group.

As each new candle completes:

- if its wick on that side qualifies and is longer than the one stored, the line moves to the new wick (new price and new start point);
- if it qualifies but is not longer, the stored line stays;
- if it does not qualify, the group ends and the line stays where it is.

If the level of an open group has already been removed (mitigated), a longer wick later in the same group starts a new level.

Upper wick CEs are bearish (resistance) and use the bear color; lower wick CEs are bullish (support) and use the bull color.

How to use it

ICT treats the 50 percent of a long wick as a level where price often reacts on a return. A weekly upper wick CE above price, for example, is a reasonable target and reaction point for a retracement into the rejected area. Grouping keeps the chart clean when several candles in a row show wicks on the same side.

With the default mitigation rule (Fully filled, wick), a lower wick CE is removed when price trades down to the wick's low, and an upper wick CE when price trades up to the wick's high. With **CE touched**, the line is removed as soon as the CE is touched.

The 1H and 4H wick CE options in the last settings section add the same kind of lines for those timeframes, using the Wick CE settings (they need **Draw wick CE** on). They use the 1H / 4H box colors and their labels start with `1H` or `4H`.

ULW CE

When a daily, weekly or monthly candle has both a qualifying upper wick and a qualifying lower wick (each at least **min wick %** of its range), a third line is drawn for that candle, labelled `ULW CE` (for example `D ULW CE`): the midpoint between the CE of its upper wick and the CE of its lower wick. It uses the wick line style and width, and the bull color when the candle closed above the line, the bear color otherwise. It is not drawn for 1H / 4H wicks. It needs **Draw wick CE** on and is controlled by **ULW CE** in the Wick CE settings. It counts against the same **keep** limit as the wick levels (a separate count per timeframe), and follows the general mitigation rule: as a single price, it is removed when price reaches it (CE touched or Fully filled by wick) or when a candle of its timeframe closes beyond it (Fully filled, close).

7. FVGs by timeframe (15m / 1H / 4H / D / W)

What you see

Each FVG of a selected timeframe is a box in that timeframe's color:

- **Outline**: solid, width 1. With the default **fill** "at margin", boxes over the candles are outlines only and boxes in the right margin are filled; "never" and "always" turn the fill off or on everywhere.
- **CE box**: a zero-height box at the midpoint, so it looks like a dotted line across the box.
- **Formation marker**: a short vertical line inside the box, from the high to the low, at the moment the FVG formed (the close of its third candle). Everything left of the marker is the pattern itself; everything right of it is time after the FVG was confirmed.
- **Label**: by default inside the box at its right end, at CE height, for example `1H FVG` or `4H IFVG`.

Inversion, the overlap rule and the VI extension work exactly as in section 4; the **FVG becomes IFVG**, **overlapping FVGs**, **extend FVG range** and IFVG / FVG label text settings of the Fair Value Gaps section apply here too. The mitigation rule of General applies to the IFVGs.

Choosing timeframes

A timeframe is drawn when **Draw FVGs of selected timeframes** is on and either:

- it is ticked and is higher than the chart timeframe; or

- **plus the next higher** is above 0 and it is one of that many listed timeframes strictly above the chart timeframe, counting from the nearest.

Ticked timeframes equal to or below the chart timeframe are ignored, so a D box is never drawn on a 1D chart. Example: on a 5m chart with nothing ticked and **plus the next higher** set to 2, the 15m and 1H FVGs are drawn; on a 1H chart the same setting draws 4H and D. This lets one setting follow you across chart timeframes.

Label layout

position	Where the label sits
Right margin	At the right end of the box, text extending to the right outside the box
Inside left	At the left edge, text extending to the right inside the box
Inside right (default)	At the right end, text extending to the left inside the box
Center	Centered between the left and right ends
Left of box	At the left edge, text extending to the left outside the box

at picks the vertical anchor: High, CE (default) or Low. The label text is built from the parts you enable: timeframe (`1H`), type (`FVG` / `IFVG`) and direction (`+` / `-`), for example `4H FVG-`. The **bold** option in the Labels section also applies here.

Right margin layout

With **Draw in the right margin** on (the default), the boxes are drawn in the right margin instead of over the candles. Each enabled timeframe gets its own column, from the lowest timeframe on the left to the highest on the right. Each FVG or IFVG is a box at its own high and low, with its CE box, and a label centered in the box: the timeframe on one line and, below it, an up arrow (bullish) or a down arrow (bearish), so the boxes can be thin. An IFVG adds an “i” to the timeframe (“1Hi”) and shows its new direction. The show TF option still applies (off leaves just the arrow); the FVG / IFVG, direction (+/-) and position options do not, and there is no formation marker.

- **offset**: bars between the last bar and the first column (default 4).
- **width**: bars per box; columns are one bar apart (default 4).
- **extend levels** (off): draws the high, the low and (with CE box on) the CE of each FVG as lines from the left edge of its box back to the point chosen next to it: **for** goes back **candles** bars before the last one (default 20), **to start** goes to the start of the FVG, and **to creation** goes to the end of the FVG’s third candle, when the gap was formed, so the third candle’s own highs and lows are not read as bounces on the FVG levels.

How to use it

The 1H and 4H boxes are the intraday PD arrays for entries and targets inside the daily range: look for price to retrace into a 1H or 4H FVG in the direction of the D/W bias, and use the CE line as the reaction level. The formation marker shows how fresh a gap is. Ticking D, W or M here draws box versions of the daily, weekly and monthly FVGs. While a timeframe is drawn as boxes (ticked, or added by **plus the next higher**, and above the chart timeframe), its FVG and IFVG lines are not drawn, since they would

repeat the boxes; its VI and wick CE lines stay. The boxes then follow the box settings, including **keep per timeframe** (6 by default, against 8 for the lines).

8. Settings reference

The tables follow the order of the settings dialog. Several controls often share one row in the dialog. A blank label means the control has no caption (it sits next to the previous one).

General

Label	Default	What it does
Daily	on	Levels from daily candles. Chart timeframe must be lower than 1D
Weekly	on	Levels from weekly candles. Chart timeframe must be lower than 1W
Monthly	on	Levels from monthly candles. Chart timeframe must be lower than 1M
extra line width per timeframe step	1 (0 to 2)	Extra width added to weekly lines (1 step) and monthly lines (2 steps) over daily ones. 0 makes all timeframes the same width. Does not affect the by-timeframe layer
remove when	Fully filled (wick)	Mitigation rule for all levels (see table below). FVGs follow the IFVG rule instead while it is enabled; IFVGs, VIs and wick CEs always follow this rule
Attenuation	0 (0 to 4)	Dims the drawings without changing their settings. 0 draws everything as configured; 4 is the most dimmed (see below)

Mitigation rules (tested on every chart bar):

Option	A bullish level is removed when	A bearish level is removed when
Never	never	never
CE touched	the bar’s low is at or below the CE	the bar’s high is at or above the CE
Fully filled (wick)	the bar’s low is at or below the level’s low	the bar’s high is at or above the level’s high
Fully filled (close)	a candle of the level’s own timeframe closes below its low	a candle of the level’s own timeframe closes above its high

For a wick CE, “low” and “high” are the ends of the wick.

Attenuation ranks (a first version, to be tuned):

Rank	Levels
important	weekly and monthly FVGs and IFVGs (lines and boxes)
normal	daily and 4H FVGs and IFVGs; weekly and monthly VIs, wick CEs and ULW CEs
minor	15m and 1H FVG boxes; daily VIs, wick CEs and ULW CEs; 1H and 4H wick CEs

Each attenuation level makes minor drawings the most transparent and important ones the least (as in ICTlevels): lines, labels and fills fade, and widths go down by 1 when dimmed enough (never below 1). In Tradovate a faded color is the color blended into the chart background: set **attbg** (white by default) to your chart background color.

Colors (by timeframe)

Label	Default	What it does
daily: bull	green	Bullish daily FVGs, IFVGs, VIs and lower wick CEs
bear	red	Bearish daily levels and upper wick CEs
weekly: bull	teal	Bullish weekly levels
bear	orange	Bearish weekly levels
monthly: bull	blue	Bullish monthly levels
bear	purple	Bearish monthly levels

Fills use these colors with the transparency set in each section.

Labels

Label	Default	What it does
Show labels	on	Labels at the right end of the D/W/M lines (and 1H/4H wick CE lines)
bold	on	Bold label text. Also applies to the by-timeframe labels
direction (+/-)	off	Appends + (bullish) or - (bearish) to FVG, IFVG and VI labels
size	normal	tiny, small, normal, large or huge, for line labels
offset	2 (min 1)	Bars between the last bar and the right end of all lines, boxes and labels
one label when hi / lo / CE closer than x daily ATR	0.05	A level whose hi, lo and CE labels would be closer than this fraction of the daily ATR(14) gets one label (`D VI`) at its CE. 0 always uses one label per line
text: daily	D	Timeframe text for daily levels
weekly	W	Timeframe text for weekly levels
monthly	M	Timeframe text for monthly levels
parts: high	hi	Text for the high line

Label	Default	What it does
low	lo	Text for the low line
CE	CE	Text for the CE line

Fair Value Gaps

Label	Default	What it does
Draw FVGs	on	D/W/M FVGs on or off
keep	8 (1 to 20)	Maximum FVGs and IFVGs kept per timeframe, both directions together. When exceeded, the oldest stored one is removed
min size	0.0	Minimum gap size in price points (before the VI extension). For NQ try 5 to 10 points to drop tiny gaps
hi/lo	solid	Style of the high and low lines: solid, dashed, dotted
(blank)	1	Base width of the FVG lines: 1, 2 or 3
CE	dotted	Style of the CE line
show: hi	on	Draw the high line and its label
lo	on	Draw the low line and its label
CE	on	Draw the CE line and its label
do not fill rectangles	off	When checked, no shaded rectangle, only the lines
transp	88 (0 to 100)	Transparency of the rectangle fill. 100 is invisible
label	FVG	Name used in FVG labels (also in by-timeframe labels)
FVG becomes IFVG on a close through it	on	Enables inversion (section 4). When off, FVGs are removed by the General mitigation rule like any other level
label	IFVG	Name used for inverted FVGs (also in by-timeframe labels)
overlapping FVGs	Most recent	Most recent, Enclosing or Keep all (section 4). Also applies to the by-timeframe layer
extend FVG range to include VIs with candle 2	on	Widens an FVG to include the body gap (VI or gap) between its middle candle and its first or third candle. Also applies to the by-timeframe layer

Volume Imbalances

Label	Default	What it does
Draw VIs	on	D/W/M volume imbalances on or off
keep	8 (1 to 20)	Maximum VIs kept per timeframe
min size	0.0	Minimum body gap in price points. Also the minimum for a VI to extend an FVG
hi/lo	solid	Style of the high and low lines
(blank)	1	Base width of the VI lines: 1, 2 or 3

Label	Default	What it does
CE	dotted	Style of the CE line
show: hi	on	Draw the high line and its label
lo	on	Draw the low line and its label
CE	on	Draw the CE line and its label
do not fill rectangles	off	When checked, no shaded rectangle
transp	88 (0 to 100)	Transparency of the rectangle fill
label	VI	Name used in VI labels

Wick CE

Label	Default	What it does
Draw wick CE	on	Wick CE levels on or off (also required for the 1H / 4H wick options)
keep	8 (1 to 20)	Maximum wick levels kept per timeframe, upper and lower together
min wick %	30.0 (1 to 100)	Minimum wick length as a percentage of the candle range for the wick to count. Higher values keep only pronounced rejection wicks
line	dashed	Style of the wick CE line
(blank)	1	Base width of the wick CE line: 1, 2 or 3
labels: upper	UW	Name used for upper wick levels
lower	LW	Name used for lower wick levels
ULW CE	on	D / W / M candles with both wicks long: a level halfway between the two wick CEs (section 6)
label	ULW	Name used for ULW CE levels

FVGs and wick CE by timeframe (15m / 1H / 4H / D / W)

Label	Default	What it does
Draw FVGs of selected timeframes	on	Turns the box layer on or off
plus the next higher	0 (0 to 6)	Also draws this many of the listed timeframes above the chart timeframe, nearest first. 0 means only the ticked ones
15m	off	15-minute FVGs
1H	on	1-hour FVGs
4H	on	4-hour FVGs
D	off	Daily FVGs as boxes
W	off	Weekly FVGs as boxes

Label	Default	What it does
M	off	Monthly FVGs as boxes
keep per timeframe	6 (1 to 20)	Maximum boxes (FVG and IFVG) kept per timeframe
min size	0.0	Minimum gap size in price points
15m: bull / bear	lime / maroon	Colors of 15m boxes
1H: bull / bear	green / red	Colors of 1H boxes and 1H wick CE lines
4H: bull / bear	teal / orange	Colors of 4H boxes and 4H wick CE lines
D: bull / bear	blue / fuchsia	Colors of D boxes
W: bull / bear	navy / purple	Colors of W boxes
M: bull / bear	olive / gray	Colors of M boxes
outline	solid	Style of the box border
(blank)	1	Border width: 1, 2 or 3 (also used for the CE box; the formation marker is one thinner when above 1)
fill	at margin	**never**: outline and CE box only. **at margin**: filled only when drawn in the right margin. **always**: filled over the candles too
transp	88 (0 to 100)	Fill transparency when the fill is on
formation marker	on	Vertical line inside the box at the close of the FVG’s last candle
CE box	on	Zero-height box (a line) at the CE
CE outline	dotted	Style of the CE box
Labels	on	Box labels on or off
size	normal	Box label size
show TF	on	Include the timeframe (15m, 1H, 4H, D, W, M) in the label
FVG / IFVG	on	Include the FVG / IFVG name
direction (+/-)	off	Include + or -
position	Inside right	Right margin, Inside left, Inside right, Center, Left of box (section 7)
at	CE	Vertical anchor of the label: High, CE or Low
wick CE: 1H	off	Adds 1H wick CE lines (needs Draw wick CE on and a chart lower than 1H)
4H	off	Adds 4H wick CE lines (needs Draw wick CE on and a chart lower than 4H)
Draw in the right margin	on	Draws the boxes in the right margin, one column per timeframe, labeled with the timeframe and an up / down arrow below it (section 7)
offset	4	Bars between the last bar and the first margin column
width	4	Bars per margin box

Label	Default	What it does
extend levels	off	Margin only: lines from the high, low and CE of each FVG back into the chart
(mode)	for	Where the extended lines start: **for** the number of candles below, **to start** the start of the FVG, **to creation** the end of its third candle
candles	20	How far back the extended lines go with **for**, in candles before the last one

The by-timeframe timeframe names in labels (15m, 1H, 4H, D, W, M) are fixed; the **text: daily / weekly / monthly** inputs do not change them.

Daily Settings

Setting	Default	What it does
Use settlement (SET) on daily	auto	Futures only (TradingView only). Which close the daily candles use for the daily FVG, VI, wick CE and ULW CE levels. **on**: the settlement price (TradingView's "use settlement as close"). **off**: the last trade. **auto**: each completed daily candle is read both ways, and the version whose largest volume imbalance or wick is bigger is used; on a tie, off. Weekly, monthly and FVGs by timeframe keep the chart's setting.
show SET label	on	On futures, shows `SET on` (orange) or `SET off` (gray) in the upper right corner, for the version used for the last daily candle.

The settlement price usually differs from the last trade of the session, so the body of a daily candle, and with it its volume imbalances and wicks, can change with that setting. In auto mode the indicator picks whichever version shows the clearer imbalance or wick for each day.

9. Alerts, limits and behaviour

Alerts

ICTcotas defines no alerts and no alert conditions. To be alerted at a level, draw a horizontal line on it or use TradingView's price alerts.

When levels are found (history window)

- Levels are detected only when a higher timeframe candle completes while the script runs through the chart's loaded bars. At each completion, the script looks at the last three completed candles of that timeframe once. It does not scan the higher timeframe history further back.

- So a 1m chart that has loaded about two weeks of bars will show daily levels from those two weeks, the weekly levels from two or three week closes, and monthly levels only if a month start falls inside the loaded history. Scroll left (to load more bars) or use a higher chart timeframe to build more history. The **keep** limits are a maximum, not a guarantee.
- Lines never start before the first loaded bar of the chart; a level from an older candle is clipped to the left edge of the history.
- For the by-timeframe layer the first candle seen on the chart is skipped, so detection starts at the second candle boundary.

Repainting and confirmation

- All levels come from completed higher timeframe candles (the data request uses offsets of one, two and three candles with lookahead), so a detected level does not repaint and does not appear before its candle closes.
- Inversion into an IFVG is checked on the completed candles of the FVG's own timeframe, so it does not repaint and is the same on any chart timeframe.
- The **Fully filled (close)** rule is checked on the completed candles of each level's own timeframe (a daily level only on a daily close), so it does not repaint and is the same on any chart timeframe.
- **CE touched** and **Fully filled (wick)** trigger as soon as the live bar's high or low reaches the level.
- Objects are drawn only on the last bar, and they are updated in place as new bars arrive.

Timeframe requirements

- Only timeframes higher than the chart are used, never the chart's own: daily levels need a chart lower than 1D, weekly lower than 1W, monthly lower than 1M, and a by-timeframe timeframe is used only when it is higher than the chart.
- When none of Daily, Weekly and Monthly is available (all unticked, or the chart timeframe is too high), the text "ICTcotas: use a chart timeframe lower than 1M" appears in the bottom-right corner.

Object limits and performance

- The script may use up to 500 lines, 500 labels and 500 boxes in total. When that is exceeded, TradingView deletes the oldest objects.
- With defaults, the D/W/M layer needs at most about 168 lines, 168 labels and 48 boxes, and the 1H and 4H boxes add 24 boxes, 12 lines and 12 labels, well within limits. Setting every **keep** to 20 with all timeframes and all kinds on can exceed 500 lines and labels.
- **keep** limits apply per kind and per timeframe, counting both directions; the oldest stored level of that kind and timeframe is removed first.
- The script makes three higher timeframe data requests (D, W, M) plus one per enabled by-timeframe timeframe. Enabling all six adds load time on long histories.

10. Troubleshooting and FAQ

No levels at all. Check that the chart timeframe is lower than 1D and that at least one of Daily, Weekly and Monthly is ticked. If the bottom-right warning is shown, one of those is the cause. Also check that the chart has enough history (section 9).

No monthly (or weekly) levels on a 1m chart. The loaded history probably does not contain a month (or week) start. Scroll left until more bars load, or open a 15m or 1H chart.

An FVG stays on the chart even though price has traded through it. With **FVG becomes IFVG on a close through it** on, an FVG is not removed by wicks; it waits for a chart candle to close beyond its far edge and then becomes an IFVG. Turn the option off if you want FVGs removed by the General rule.

A level disappeared without being touched. The **keep** limit removed the oldest level of that kind and timeframe, or a newer overlapping FVG replaced it under the **Most recent** or **Enclosing** overlap rule.

A wick CE line jumped to a new price. A later candle in the same wick group had a longer wick on the same side; the line moves to the longest one.

My 15m boxes do not appear. The 15m timeframe is ignored on charts above 15m. Also check **Draw FVGs of selected timeframes**.

I see the daily FVGs twice. The D option in the by-timeframe section draws daily FVGs as boxes on top of the daily lines. Untick it, or untick **Draw FVGs** in the Fair Value Gaps section.

IFVG results differ between chart timeframes. Inversion is tested on chart-bar closes, so a 1m close through a level inverts it on a 1m chart while a 15m chart may not see a 15m close beyond it.

Labels overlap the price bars. Increase **offset** in the Labels section, reduce label **size**, or switch the box labels to **Right margin**.

Changing the time zone changed nothing. Correct: levels follow the exchange’s daily, weekly and monthly candles, not the chart time zone.

11. Glossary

Term	Meaning
PD array	Price delivery array: any reference area (FVG, VI, wick, order block) where price is expected to react
FVG	Fair Value Gap: three-candle pattern where candle 1 and candle 3 wicks do not overlap
IFVG	Inversion FVG: an FVG that price closed through; it now acts in the opposite direction
VI	Volume imbalance: gap between the bodies of two consecutive candles whose wicks still overlap
CE	Consequent encroachment: the 50 percent midpoint of a gap or wick
Wick CE	Midpoint of a candle wick; here the longest wick of a run of same-side wicks
UW / LW	Upper wick / lower wick
ULW	Midpoint between the upper and lower wick CEs of one D / W / M candle
Mitigation	Price returning to a level and using it up; the level is then removed

Term	Meaning
Bullish / bearish level	Support below price / resistance above price
Formation marker	Vertical line in a by-timeframe box at the close of the FVG’s third candle
Draw on liquidity	The level price is expected to move toward next
HTF	Higher timeframe